Comparison Between Java, C#, and PHP

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Unique features of each style guide

1-Java – Google Java Style Guide

Unique idea: It is strict, detailed, and designed to be automatically enforceable

- Focus on hard rules that MUST be followed (not just suggestions)
- Very precise formatting system, like:
  - space indentation
  - max line length = 100 characters
- Enforces exact file structure order (package → imports → class)
- Avoids flexibility → goal is uniform code everywhere

2- C# – Microsoft Coding Conventions

Unique idea: - It combines style + best practices + tool support (not just formatting**)**

- Not just formatting → includes how to write good code logic (LINQ, async/await, exceptions)
- Encourages using modern language features and avoiding outdated ones
- Integrated with tools:
  - Visual Studio
  - code analyzers
- Flexible → teams can adapt to rules

3- PHP – PSR-2

Unique idea: It focuses on making code from different developers look the same

- Created by PHP-FIG (community), not a company
- Main goal: reduce differences between projects
- Uses shared rules across frameworks
- Defines clear formatting like:
  - 4 spaces indentation
  - consistent braces placement

Common features between the three

All three guides share the same core goals:

1. Improve readability

- All aims to make code easier to read and understand
- PSR-2 explicitly says it reduces “cognitive friction”

2. Ensure consistency

- Same structure and formatting across teams
- Microsoft also emphasizes consistency in all samples

3. Define coding rules

They all include rules for:

- Naming conventions
- Indentation & spacing
- File / code structure

4. Support teamwork

- All are designed to make collaboration easier between developers